FieldBrain

Technician Manual Q&A with Honest Hand-Off

AMD Developer Hackathon — ACT III · Track 1: Intelligent Industry

Submission build: offline demo mode — local BM25 retrieval, no external model call.

github.com/QuincyGB/fieldbrain

BEAT 1 — THE PROBLEM

The documents exist. Nobody can query them from a rooftop.

- Field technicians lose time digging through equipment manuals and old work orders.
- Junior techs miss the recommended next action — the answer is in the manual, but not at the moment of work.
- Built by a practitioner with 10 years managing residential and commercial service accounts — the workflow is the one he lived.

BEAT 2 — GROUNDED ANSWERS

Answers grounded in the tech's own manuals — with named sources

- FieldBrain indexes uploaded manuals and maintenance logs; retrieval and answer assembly are local (BM25 over the indexed corpus).
- Demo Q1: "compressor short-cycling, fault code E47 — what do I check first?"
- Answer retrieved from the RTU-9000 manual's E47 section, corroborated by the 2026-03-14 maintenance log — every claim cites document + section.

BEAT 3 — THE MANUAL'S OWN CHECKLIST

The recommended actions are the manual's own procedure, in order

- Alongside the answer, FieldBrain returns a recommended action checklist extracted from the manual's own procedure — not generated advice.
- For E47: inspect and wash the condenser coil, check filters and the supply-fan belt, watch one full cycle, then gauge pressures at cut-out.
- The tech gets the next step, in sequence, traceable to the source document.

BEAT 4 — THE DIFFERENTIATOR

When the docs can't answer, FieldBrain hands off — it never guesses

- Demo Q2 asks for the refrigerant type and factory charge weight. The documents deliberately never state them.
- Retrieval confidence falls below the gate (best match 5.906, term coverage 0.532), so no answer is invented.
- Instead: a structured hand-off summary — question, what was searched, closest matches, suggested route — for a senior tech or dispatcher.

BEAT 5 — TARGET ARCHITECTURE

Same API, same behaviour — production runs on AMD

- The demo build runs in offline demo mode: local BM25 retrieval, so it works with no external model call.
- Production target: an open model served by vLLM on AMD Developer Cloud (ROCm), behind the same API.
- The grounding, checklist, and hand-off behaviour are identical in both modes — they live in the retrieval layer, not the model.
- What changes when the credits activate: the answer-generation backend. What doesn't: the honesty guarantees.

Stack: FastAPI · BM25 · RAG · Docker · Python · vLLM (target) · ROCm (target) · AMD Developer Cloud (target)